

IRRBB Advanced Modeling Concepts

Comprehensive Guide with Conceptual and Quantitative Details

1. Modeling Prepayment Risk in IRRBB

Conceptual Foundations

Prepayment risk is critical in IRRBB, especially for banks with mortgage portfolios. Lower interest rates increase borrower refinancing, leading to:

- Shortened asset duration
- Reinvestment risk
- Non-linear cash flow sensitivity

Key Regulatory References:

- Basel Committee (BCBS 368): Behavioral option modeling
- OCC 2011-12: Assumptions on loan prepayments

Quantitative Modeling Approaches

1. Public Securities Association (PSA) Model:

100% PSA = 0.2% CPR (month 1), rising 0.2%/mo to 6% at month 30.

Banks scale (e.g., 150% PSA).

```
def psa_model(months, psa_speed=100):  
  
    cpr = [min(0.002 * psa_speed / 100 * min(m, 30), 0.06) for m in  
months]  
    return [1 - (1 - cp)**(1/12) for cp in cpr] # Monthly prepay rate
```

2. Richard and Roll (1989) Model:

Considers:

- Refinancing incentive
- Seasoning
- Seasonality
- Burnout

$CPR = RI \times \text{Seasoning} \times \text{Seasonality} \times \text{Burnout}$

Hands-on Exercise

- Mortgage pool: \$100M principal, 5% coupon, 30-year maturity
- Model prepayments using PSA and Richard-Roll
- Compare EVE impact under -200bps rate shock

2. Deposit Beta Estimation Techniques

Conceptual Framework

Deposit betas measure how deposit rates respond to market rate changes:

$$\beta = \Delta \text{Deposit Rate} / \Delta \text{Market Rate}$$

Low beta (0.1–0.3): Core deposits

High beta (0.7–1.0): CDs, money markets

Regulatory Requirements

- EBA: Justify beta assumptions
- Fed SR 15-18: Empirical estimation required

Quantitative Estimation Methods

1. Historical Regression Approach:

$$\Delta D_t = \alpha + \beta \Delta M_t + \varepsilon_t$$

Steps:

- Collect 5+ years of monthly data
- Run regressions
- Segment by product

2. Cointegration Analysis (Error-correction model):

$$D_t = \mu + \beta M_t + \gamma(D_{t-1} - \beta M_{t-1}) + \varepsilon_t$$

3. Forward-Looking Survey Approach:

- Survey depositors
- Combine with historical behavior

Hands-on Exercise

- Load Fed & bank deposit rate data (5 years)
- Estimate betas with OLS
- Simulate NII impact (+100bps shock)

3. Regulatory Capital Implications

EVE Shock Scenarios

Basel III standard shocks:

- Parallel up/down (+200/-200bps)
- Steepener, Flattener
- Short rates up/down

Capital Calculation

$$\text{Capital Charge} = \max(\text{EVE_shocked} - \text{EVE_base}, 0) \times \text{Supervisory Factor}$$

Behavioral Adjustment Framework

Banks must document:

- Prepayment assumptions
- Deposit decay: Survival analysis
- Deposit betas: Empirical evidence

Implementation Case Study

Scenario:

- \$5B mortgage portfolio (4%, 10yr avg life)
- \$3B non-maturity deposits (0.5%)

Steps:

- Model prepayments (Richard-Roll)
- Estimate deposit betas
- Calculate EVE under Basel shocks
- Check capital requirement triggers

Tools Required

Python/R, ALM software (QRM/BancWare), Bloomberg/Refinitiv

Further Reading

- Basel Committee (2016) – IRRBB
- OCC (2020) – IRRBB Handbook
- Fabozzi (2016) – Mortgage-Backed Securities
- Jarrow & van Deventer (1998) – Valuation of Demand Deposits